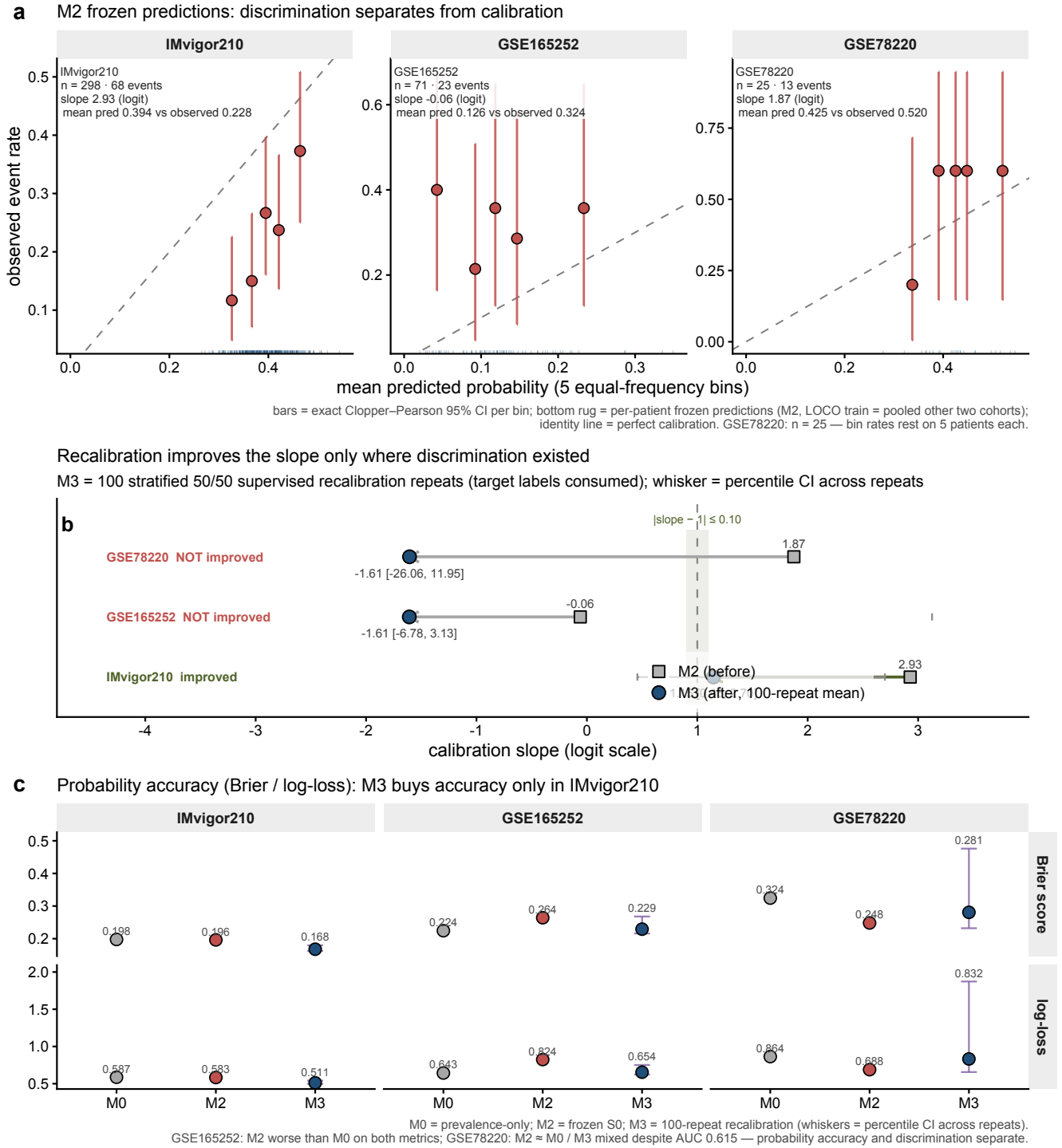

a
M2 frozen predictions: discrimination separates from calibration
IMvigor210
GSE165252
GSE78220
IMvigor210
n = 298 · 68 events
slope 2.93 (logit)
mean pred 0.394 vs observed 0.228
GSE165252
n = 71 · 23 events
slope -0.06 (logit)
mean pred 0.126 vs observed 0.324
GSE78220
n = 25 · 13 events
slope 1.87 (logit)
mean pred 0.425 vs observed 0.520
observed event rate
mean predicted probability (5 equal-frequency bins)
bars = exact Clopper–Pearson 95% CI per bin; bottom rug = per-patient frozen predictions (M2, LOCO train = pooled other two cohorts);
identity line = perfect calibration. GSE78220: n = 25 — bin rates rest on 5 patients each.
Recalibration improves the slope only where discrimination existed
M3 = 100 stratified 50/50 supervised recalibration repeats (target labels consumed); whisker = percentile CI across repeats
b
|slope − 1| ≤ 0.10
GSE78220 NOT improved
1.87
-1.61 [-26.06, 11.95]
GSE165252 NOT improved
-0.06
-1.61 [-6.78, 3.13]
IMvigor210 improved
2.93
M2 (before)
M3 (after, 100-repeat mean)
calibration slope (logit scale)
c
Probability accuracy (Brier / log-loss): M3 buys accuracy only in IMvigor210
IMvigor210
GSE165252
GSE78220
Brier score
log-loss
0.198
0.196
0.168
0.224
0.264
0.229
0.324
0.248
0.281
0.587
0.583
0.511
0.643
0.824
0.654
0.864
0.688
0.832
M0
M2
M3
M0 = prevalence-only; M2 = frozen S0; M3 = 100-repeat recalibration (whiskers = percentile CI across repeats).
GSE165252: M2 worse than M0 on both metrics; GSE78220: M2 ≈ M0 / M3 mixed despite AUC 0.615 — probability accuracy and discrimination separate.